Siyuan Feng

BSc Mathematics, University of Warwick | Machine Learning Theory
Email: u5605586@live.warwick.ac.uk | Phone: +44 7303 077764

EDUCATION

University of Warwick Sep 2024 – Jun 2027 (expected)
BSc Mathematics
- Year 1 average: 82.5%; Year 2 average: 82.0% (First-Class level; estimated US GPA: 3.94/4.00).

RESEARCH INTERESTS

Machine Learning Theory: Mathematical foundations of neural-network training, with a focus on learning-rate scaling and transfer across model widths and training budgets, as well as checkpoint averaging.

RESEARCH EXPERIENCE

Learning-Rate Scheduling and Neural-Network Training Jul 2026 – Present
Research with Prof. Fanghui Liu
- Investigating the mathematical foundations of learning-rate scheduling, focusing on how learning rates vary during training and influence neural-network optimisation.

Checkpoint Averaging and Learning-Rate Decay Oct 2026 – Present
Research with Prof. Ranko Lazić
- Investigating when checkpoint averaging can approximate learning-rate-decayed training, and how to improve its accuracy without increasing computational complexity.
- Identified theoretical and experimental limitations of QAM checkpoint averaging and derived a local loss-gap expansion relative to training with explicit learning-rate decay.

Two-Step Optimal Learning-Rate Geometry under μP May 2026 – Jul 2026
Warwick URSS | Supervised by Prof. Fanghui Liu
- Investigated the mathematical basis of transferring learning rates from smaller to wider neural networks, with the aim of reducing repeated hyperparameter tuning.
- Analysed a two-step linear-network model, establishing conditions for a unique optimal learning rate and constructing examples with several equally optimal choices.

SELECTED ACADEMIC PROJECTS

Quantum Computing and Variational Algorithms Jan 2026
- Implemented quantum algorithms using MindQuantum, including quantum Fourier transform (QFT), QAOA for Max-Cut, and VQE for molecular energy estimation; conducted numerical simulations of variational optimisation methods.

Mathematical Computing and Machine Learning Jan 2025 – Mar 2025
University of Warwick – Mathematics by Computer | Supervised by Prof. Damiano Testa
- Implemented and evaluated regression models and neural networks using Python and NumPy, investigating gradient-based optimisation, regularisation, convergence, and generalisation through numerical experiments.

INTERNSHIP

University of Warwick – Undergraduate Mathematics Supervisor Oct 2026 – Present
- Employed by the University of Warwick to lead first-year mathematics tutorials, prepare materials, mark assignments, and provide individual feedback.

Tianma Microelectronics Co., Ltd. – Algorithm Engineering Intern Aug 2025 – Sep 2025
- Built a pixel-level E-paper video ghosting simulation pipeline in MATLAB and Python, covering frame extraction, drive mapping, and grayscale output.
- Replaced pixel-wise loops with NumPy vectorization in a history-frame-difference algorithm, reducing single-run simulation time by approximately 85%.

OUTREACH EXPERIENCE

Mathematics Content Creator, Bilibili Ongoing
- Created 60+ original videos explaining calculus and advanced integration problems, reaching 12,000+ followers and 500,000+ cumulative views. Developed accessible problem-solving explanations for a broad online mathematics audience.

AWARDS & RESEARCH FUNDING

University of Warwick URSS Research Bursary — £500 2026
Awarded in support of undergraduate mathematical research.

2025 Quant Championship — 4th out of 200 participants 2025
Competition covering statistics, computational problem-solving, and numerical analysis; awarded a £75 prize.

TECHNICAL SKILLS

- **Scientific computing:** Python (PyTorch: neural-network training; NumPy/SciPy: simulations and learning-rate sweeps; pandas/Matplotlib: data analysis and plots); MATLAB: algorithm simulation.
- **High-performance computing:** Sulis HPC, Linux/SSH, Slurm (GPU job submission and monitoring), PyTorch GPU experiments.
- **Research workflow:** Git (version control), Jupyter, Conda (environments), and LaTeX (mathematical papers).